Phase 1 — Clean base model

Phase 1 turns the current static FreeCAD file (1,764 objects, no expressions) into a regenerated, parametric, human-readable assembly. Phase 2 (cap redesign) starts only when every check below passes.

Deliverables

1. `build_assembly.py` and its pieces — run by Jordane as FreeCAD macros on his Mac; they create the clean document in his FreeCAD, which he saves as `keycaps_test_assembly_clean.FCStd`.
2. A printed check report (numbers, pass/fail) each run, pasted back to chat; the visual check happens in the same FreeCAD window.

How the code is written

- One piece at a time, each reviewed in chat before the next; a step that grows big moves to its own file and is imported.
- Naming: `verb_noun`, one noun per function — `read_` reads a file, `make_` creates FreeCAD objects, `place_` / `seat_` position things, `check` verifies, `finish` cleans up. Step numbers live in comments, not names, so reordering never forces renames.
- Each function has a docstring saying what goes in and what comes out; constants named, not inlined; no CLI flags, no project scanning; file paths are Jordane's Mac paths.
- Conventions carried over from the sensor model (`fc_helpers.py`): VarSet with inputs as numbers and derived values as expressions on the VarSet; every shape a Part primitive bound by expressions (`bind` / `box` / `cylinder`); tools fused then `Part::Cut`; one `App::Part` container per component holding its helpers; only finished bodies visible (`show_only`); colors through `ViewObject` only when the GUI is up (`paint`); datasheet-explicit values stay literal, everything else relative.
- Every piece runs as a FreeCAD macro (`__main__` block builds into a new document and leaves it open). `fc_helpers.py` and `sensor_sk8707.py` sit in the Macro folder next to it so `import` works.
- Running the generator always rebuilds the whole clean document from the two inputs (`.kicad_pcb`, original FCStd). Nothing is hand-placed.

- Every VarSet property carries a plain-language tooltip saying what the value means (derived ones also show their formula), and containers, library bodies and key objects get a description (Label2, the tree's Description column). Parameter tables in code are `name: (value, tooltip)`.

Steps

Each step is one function in `build_assembly.py` (skeleton written 2026-10-09); shared FreeCAD helpers live in `fc_helpers.py`. Status: sensor done; the rest are empty and filled in this order.

#	Function	Why it is needed	Status
1	`get_pcb_config(path: str) -> PcbConfig`	Positions, pads, holes, outline and model paths must come from the board file, not be typed in; plain-Python s-expression parse, KiCad Y flipped, origin at REF**. Pad rotation in the file is absolute (verified against KiCad's exported pad solids), stored relative to the footprint `**`	done
2	`build_3d_model_library(doc: Document, pcb: PcbConfig) -> Library`	Each distinct model file named by the footprints is imported once into a hidden Library group. STEP/IGES go through FreeCAD's own importer (`ImportGui` with the GUI), which keeps the file's parts, their names and colors, e.g. Cherry's switch arrives as one App::Part holding SOCKET, WING ×2, AKTUATOR, TENSION_SPRING; BREP is read as one shape, STL/OBJ as meshes; VRML unsupported. Everything else Links to these. KiCad's ${VAR} paths are expanded from KICAD_PATH_VARS; missing files are reported and skipped	done

#	Function	Why it is needed	Status
3	`build_root_varset(doc: Document, pcb: PcbConfig) -> Object`	Root VarSet `Params.board_thickness` comes from the board file; inputs `cu_thickness 0.04`, `solder_sink 0.02`, `keycap_ceiling 3.5`, `keycap_seat 0`, `cap_offset 0`; derived `board_top`, `pad_top`, `part_seat`, `back_pad_bottom`, `keycap_ceiling_z`, `sensor_top_z` (= back_pad_bottom + solder_sink: the sensor is soldered to the back pads; the assembly has no screws, so no screw-head gap)	done
4	`make_pcb(doc: Document, pcb: PcbConfig) -> Object`	Board box cut by `<REF> Mount Holes` groups that follow each component's `<REF> Params`. The box is sized by `Layout.board_left/right/down/up`: on each side `max(KiCad's edge, outermost switch courtyard + KiCad's original margin)`, so it grows and shrinks with the switches but never below KiCad's size (tested: gap 3.05 → KiCad's 47.0 × 47.6, gap 8 → 52.0 × 52.0, gap 1 → back to 47.0 × 47.6)	done
5	`make_solder_pads(doc: Document, footprint: Footprint) -> Object`	Pad solids from the footprint definitions, `cu_thickness` tall, on the board face (rebuilt, not sorted from the explode)	done

#	Function	Why it is needed	Status
6	`group_by_component(doc: Document, footprint: Footprint, pcb: PcbConfig, library: Library) -> Object`	`<REF>` container (returned, an `App::Part`) holding `<REF> Params`, `<REF> Pads`, the body Link on `part_seat`, for switches the keycap Link (on the board top until step 8) and the nested LED## / D## containers. Links are placed by `place_model(doc: Document, footprint: Footprint, model: Model, library: Library, base_z: str) -> Object | None` with KiCad's own model offset and rotation, read into `Model` by step 1	done
7	`place_trackpoint_sensor(doc: Document, footprint: Footprint, library: Library) -> Object`	`REF**` container: `REF** Params`, `REF** Pads`, the parametric sensor without its screws (build_sensor(doc, with_screws=False)) from `sensor_sk8707.build_sensor` hanging under the board (top face at `sensor_top_z`, stem up through the ×7.5 hole, turned with the footprint), and the cap Link with its socket floor on the stem top + `cap_offset`. The floor is found by `_socket_floor_z(shape, x: float, y: float) -> float` (lowest material on the cap's axis). The vendor SK8707 STEP is not used	done
8	`seat_keycaps(doc: Document, pcb: PcbConfig) -> dict[str, float]`	Moves each keycap so its inner ceiling (underside of the top plate, probed 3 mm from the center to miss the center nub, by `_keycap_ceiling_z(shape, x: float, y: float) -> float`) sits at `keycap_ceiling_z` = `part_seat` + 3.5 + `keycap_seat`, measured from where the switch sits, as in Cherry's files	done

#	Function	Why it is needed	Status
9	`check(doc: Document, pcb: PcbConfig) -> bool`	Printed pass/fail report (28 checks): board volume, pads, parts sunk 0.020 into pads, 12 pins in holes, keycaps clear of switches, stem through the hole, cap floor on the stem top, cap clear of keycaps. Phase 2 gaps via `_clearance(a, b) -> float` (exact) only when `MEASURE_GAPS = True` (a few minutes). Sandbox run 2026-10-10: 27/28 pass in 95 s; the fail is SW33's mnhttn clip hooks overlapping Cherry's arm ends (0.006 mm³, a phase 2 finding)	done
10	`finish(doc: Document) -> None`	Hides every helper shape (anything a cut, fuse, extrusion or offset is built from) and fits the view. Colors are already set as each step makes its bodies; library models stay visible because the Links show them	done

All functions are type-hinted. Data read from the board is carried in dataclasses: `Pad` (number, shape, x, y, rot, width, height, layer, corner_ratio), `Hole` (x, y, diameter), `Model` (path, offset, rotate: KiCad's own 3D placement), `Footprint` (ref, x, y, rot, pads, holes, models), `PcbConfig` (thickness, outline, footprints by ref). `Library` is `dict[str, Object]` keyed by model file name without extension. FreeCAD types are aliased in `fc_helpers.py` as `Document`, `Object` and `RGB`. Helpers: `make_component_varset(doc: Document, footprint: Footprint) -> Object`, `_on_board(params: str, x: float, y: float) -> tuple[str, str]`, `make_mount_holes(doc: Document, footprint: Footprint) -> Object`, `_pad_size(pad: Pad) -> tuple[float, float]`, `_rounded_rect(doc: Document, name: str, w: float, h: float, r: float, x: float, y: float, z: str) -> Object`, `_model_placement(params: str, model: Model, base_z: str) -> str`, `place_model(doc: Document, footprint: Footprint, model: Model, library: Library, base_z: str) -> Object | None`. Colors: our own bodies are painted as each step makes them (`PCB_GREEN`, `COPPER_GOLD`, the sensor's own palette); vendor models keep their STEP colors.

Keycap seating (measured 2026-10-09)

The MX ULP keycap is a 0.4 mm shell with a 1.4 mm skirt that snaps onto the ends of the switch's butterfly arms at its four corners; there is no stem socket. Seat height was found by lowering Cherry's official keycap model onto Cherry's official switch assembly and looking for the one height where the arms neither pierce the shell nor collide with the clip hooks.

Reference (Y-up in Cherry's files; PCB top = 0)	Height
Switch frame top	1.60 mm
Switch actuator cam top	2.86 mm
Switch arm ends (highest point of the bare switch)	3.47 mm
Keycap cavity ceiling at rest	3.50 mm
Keycap top surface (0.4 mm shell)	3.90 mm
Keycap skirt bottom	2.10 mm

Rule in the generator: `keycap ceiling z = board_top + 3.5 mm + keycap_seat` (`keycap_seat` default 0). For the dactyl and mnhttn caps the same lowering test is run against the switch model and printed as a check, since both copy Cherry's clip geometry. The 4.5 mm / 4.0 mm offsets in the KiCad footprint were estimates and are replaced by this rule.

Sources: `CHERRY_MX_ULP_KeyCap.stp`, `CHERRY_MX_ULP_mx6c-t3nb_customer_00_asm.stp` and `CHERRY_MX_ULP_KeycapMeasures.pdf` in frankbenoit/Cherry_MX_ULP.

Hierarchy

```
Params (VarSet)   board_thickness 1.6 · cu_thickness 0.04 · solder_sink 0.02
· keycap_seat 0
                  sensor_* (see Sensor tab) · cap_offset 0
Library (hidden)  LIB Switch Cherry MX6C-K3NB · LIB Keycap ulp dactyl · LIB
Keycap mnhttn
                  LIB Cap Sprintek 109-00060 · LIB LED LTST-FC12WEGBD-5A ·
LIB Diode SOD-523
PCB
  Board                       box from outline × board_thickness, cut by:
  SW33 Mount Holes … SW48     3 cylinders each, follow SW## Params
  REF** Mount Hole            Ø7.5 at the sensor center (assembly origin)
SW33  (SW33 Params: x, y, rot)
  SW33 Switch                 z = board_top + cu − solder_sink
  SW33 Keycap                 socket on stem, + keycap_seat
  SW33 Pads                   7 pads from the footprint
  LED33 → LED33 Body, LED33 Pads     (offset relative to SW33)
  D33   → D33 Body,   D33 Pads
SW34 … SW47 … SW48            same pattern
REF**
  REF** Sensor                placeholder: sensor PCB box + 2.4 mm square
stem, under the board
  REF** Cap                   on the stem + cap_offset
  REF** Pads                  4 roundrect pads, back side
```

Naming

Object	Label pattern	Example
Component container	`<REF>`	`SW33`, `LED33`, `D33`, `REF**`
Per-component parameters	`<REF> Params`	`SW33 Params`
Body instance	`<REF> <Role>`	`SW33 Switch`, `SW33 Keycap`, `LED33 Body`, `REF** Cap`
Pads group	`<REF> Pads`	`D33 Pads`
Hole group	`<REF> Mount Holes`	`SW33 Mount Holes`

Object	Label pattern	Example
Library master	`LIB <Part name>`	`LIB Keycap ulp dactyl`

Checks (printed every run)

Check	Pass condition
Rebuilt pads vs KiCad's exported pad solids	max XY/Z deviation < 0.01 mm (also settles KiCad's pad-rotation convention)
Pad bottom on board top	gap = 0.000 mm
Component into pad	overlap = 0.020 mm
Switch pins in holes	all 12 pins: clearance to hole wall ⩾ 0
Keycap on switch	keycap ∩ switch volume = 0; stem top inside socket
Sensor stem through hole	stem ∩ board = 0; stem top = board_top + 0.8 mm (see Sensor tab)
Cap on stem	cap ∩ keycaps = 0; socket floor on stem top
Gap baseline for phase 2	min XY distance cap↔each keycap and keycap↔keycap, reported in mm

Current state for reference: keycap inner edges sit 3.0–3.3 mm from center and the cap radius is 4.5 mm, so in plan view the cap already overlaps the keycap corner zone; only rounded corners keep them apart.

Found while writing steps 8–9 (2026-10-10): the dactyl keycap and the cap STEP files are 12,000-triangle meshes, and the dactyl is a closed skin, not a solid, so step 2 now wraps closed skins into solids. KiCad's model z offsets are ignored (the LED's −0.01 sank it 0.03); every height comes from Params. Step 2 skips the vendor SK8707 STEP (its "Sensor Stem" label collided with ours). The mnhttn keycap's four clip hooks reach about 0.2 mm into Cherry's arm ends; raising the keycap does not clear them.

Layout parameters (added 2026-10-10): `make_layout_varsets(doc: Document, pcb: PcbConfig) -> Object` runs in step 3 and makes the `Layout` VarSet: `switch_gap` (courtyard edge to courtyard edge, the same left-right and top-bottom, centered on the sensor; starts at the board's smallest gap, 3.05), `sensor_hole_d` (starts at the board's 7.5), and the switch courtyard reach read from the footprint (`crtyd_left/right/down/up`, stored positive because a Length VarSet value cannot be negative). Switch Params x/y follow `Layout.sw_x_*` / `sw_y_*`; LED and diode Params follow their switch with KiCad's offset; the sensor hole cutter radius is `Layout.sensor_hole_d / 2`. The switch footprint's big courtyard was centered to ±9.5 in the board (was −10..9). Step 9 adds: equal courtyard gaps, courtyards inside the board, sensor hole wall ⩾ 0.25 mm (KiCad min_hole_clearance) from switch pads/holes, cap clear of the board. First result: at gap 3.05 the cap hits the three dactyl keycaps (the KiCad layout had 3.61 top-bottom), so with the current cap and keycaps the equal gap must stay between 3.05 and 3.61; finding the exact limit is phase 2 work.

Switch footprint vs Cherry's drawing (checked 2026-10-10)

The board's `Cherry:Cherry_ULP_SMD` footprint is identical to `Cherry_ULP.pretty/Cherry_ULP_SMD.kicad_mod` in [frankbenoit/Cherry_MX_ULP] (community-made). It was compared with Cherry's own `CHERRY_MX_ULP_PCB_Layout.pdf` in the same repo, measured from the PDF's vector lines (5:1 drawing, scale confirmed by the 11.6 mm hole spacing and both hole diameters). Values below in the KiCad footprint frame (Y down).

Feature	Footprint	Cherry drawing	Difference
NPTH Ø1.05	(−5.8, 1.2)	(−5.8, 1.2)	none
NPTH Ø1.2 ×2	(5.8, ±3.26)	(5.8, ±3.26)	none
Pad 1 (0.7 × 3.4)	x −0.65	x −0.60	0.05 mm
Pad 2 (0.7 × 3.4)	(1.8, 2.3)	(1.8, 2.3)	none
Pad 3, left upper	(−6.2, −3.9), 1.6 × 3.8	(−6.2, −4.1), 1.6 × 3.8	0.2 mm
Pad 3, left lower	(−6.2, 4.5), 1.6 × 3.0	same	none
Pad 3, right outer ×2	(6.2, ±5.025), 1.6 × 1.8	(6.2, ±5.075), 1.6 × 1.85	0.05 mm

Feature	Footprint	Cherry drawing	Difference
Pad 3, right middle	(6.2, 0), 1.6 × 2.6	same	none
LED window	LED48 at (−0.25, −3.0) from its switch	LED at (0, −2.8), body ⩽ 0.8 mm tall	0.25 / 0.2 mm

Cherry's official switch model (`CHERRY_MX_ULP_mx6c-t3nb_customer_00_asm.stp`, Y-up, plan view X–Z mirrored in X) puts its three locating pins (0.93 mm square at 0.2 mm below the PCB) at (−5.8, 1.2) and (5.8, ±3.2), inside the holes. That model does not include the solder terminals, so the drawing is the only source for the pads. The repo README says its own `Cherry_MX_ULP.step` is for visualization only.

Why they differ: the community footprint dates from 2023 (pashutk, "Tune offsets"), and Cherry's drawing was only added to the repo on 2025-10-29 ("files received from Cherry"). The keep-out zone was also 0.05 mm off (Cherry: x −5.05 to −1.25, y 2.2 to 4.2). **Corrected `Cherry_ULP_SMD.kicad_mod` written 2026-10-10**: pad 1, the three off pads and the keep-out zone now match the drawing; all 10 pads and holes verified by script against the measured values.

Switch 3D models compared (2026-10-10)

Both files are byte-identical to copies in frankbenoit/Cherry_MX_ULP. **Cherry's**: `CHERRY_MX_ULP_mx6c-t3nb_customer_00_asm.stp`, made by E. Ruff in Creo on 2023-01-17, added to the repo as "files received from Cherry". **Community**: `Cherry_MX_ULP.step` ("ulp switch model v8"), exported from Autodesk on 2023-07-21 by pashutk; the repo README calls it visualization-only. Compared in the board frame: Cherry's placed so its pins sit in the footprint holes, the community one placed exactly as the footprint places it (rotate Z −90 stored, offset −5.5, 0, 3).

	Cherry (T3NB)	Community
Parts	5, named: SOCKET, WING ×2, AKTUATOR, TENSION_SPRING	8, unnamed (Component1–5, SOLID); housing split in two
Locating pins	round, Ø0.95 at the root, Ø0.75 tip, 0.38 deep; exactly at (−5.8, −1.2), (5.8, ±3.2); clearance in holes +0.05 / +0.065	straight cylinders the same size as the holes, 0.40 deep, 0.045 mm off center, so they cut into the board by 0.045

	Cherry (T3NB)	Community
Arm ends (keycap clips)	Y ±6.54, X to ±6.68, top Z 3.475	Y ±6.35, X to ±6.88, top Z 3.53; shape differs up to 0.51 mm
Housing top	1.65	1.70 / 1.80
Overall		98% of each surface within 0.3 mm of the other, about half within 0.05 mm; Cherry's housing interior up to 0.85 mm different
Material volume	156 mm^3	95 mm^3

Consequence: the keycap ceiling (3.50) was measured on Cherry's model; the community arms (3.53) would poke 0.03 mm into the keycap. Both models have small built-in overlaps between their own parts (largest: Cherry actuator into socket, 0.94 mm^3), so collision checks should treat the switch as one unit. Datasheets: MX6C-K3NB = Click, MX6C-T3NB = Tactile, same 1.8 mm travel, 0.8 mm pretravel, forces.

Cost

About 5–8 turns, roughly 40–60k tokens, most of it the script itself. The 60 MB STEP is never read.